THE UNITED NATIONS COMMITTEE ON THE PEACEFUL USES OF OUTER SPACE: PAST ACCOMPLISHMENTS AND FUTURE CHALLENGES

*S. Neil Hosenball**

I. Background

The United Nations Ad Hoc Committee on the Peaceful Uses of Outer Space came into being on December 13, 1958,[1] one year after the launch of Sputnik and almost simultaneous with the birth of NASA.[2] A part of its charter was to explore "the nature of legal problems which may arise in the carrying out of programs to explore outer space."[3] One year later, the Ad Hoc Committee was transformed into a standing committee of the United Nations. The Committee, which initially consisted of representatives from 24 countries, was increased in 1961 to 28, in 1974 to 37, and today, after the addition of 10 new members in 1977, has 47 member nations participating. The member nations on the Committee mirror the UN membership as a whole, having African, Asian, Latin American, Western European, Eastern European, Mid East, First World countries, Third World countries, and countries in every stage of development.[4]

These recent increases in membership evidence the ever growing international interest in the purposes for which outer space is explored and used and in the achievement of an orderly basis for the conduct of space activities. They also demonstrate that many nations are eager to participate directly through international cooperation in various aspects of outer space in such applications as those involved in scientific research, weather forecasting, communications, and remote sensing.

The Committee has a permanent UN staff in the Outer Space Division of the UN Secretariat, which includes a space application expert whose prime role is to facilitate the exchange of scientific and technical information among member States of the United Nations and, in particular, to assist the developing countries so that they may become familiar with and participate in space applications that can serve their special interests and needs.

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*General Counsel, National Aeronautics and Space Administration; U.S. Representative to the 1979 Session of the U.N. Committee on the Peaceful Uses of Outer Space.

[1]For discussions of the Ad Hoc Committee, *see* Jessup and Taubenfeld, The Ad Hoc Committee on the Peaceful Uses of Outer Space, 53 Am. J. Int'l. 877 (1959).

[2]National Aeronautics and Space Act,P.L. 85-568, 42 U.S.C. 2451 *et seq.*

[3]*Supra* note 1.

[4]U.N. Doc. A/AC.105/218.

The Committee established two subcommittees, a Legal Subcommittee and a Scientific and Technical Subcommittee. The Legal Subcommittee holds a four-week session in the spring of each year, and the Scientific and Technical Subcommittee also meets in the spring for a two or three-week session, usually prior to the Legal Subcommittee meeting. Each subcommittee at the conclusion of its session prepares a report to the parent Committee on the Peaceful Uses of Outer Space (COPUOS) to be considered during a two or three-week session in late spring or early summer. Similarly, the parent committee at the end of its session issues a report for discussion in the General Assembly's Special Political Committee. The Special Political Committee then drafts a resolution for submission to and subsequent adoption by the General Assembly.

II. Consensus Procedure

All members of the committee on the Peaceful Uses of Outer Space are represented on its two subcommittees. Generally, the delegates to the Legal Subcommittee are lawyers or diplomats, whereas most delegates to the Scientific and Technical Subcommittee have a scientific or technical background. The Committee and its subcommittees work on the basis of consensus. The alternatives to operating by consensus would be to decide issues by majority vote. The Committee recognized at an early stage of its history that dealing as it would be in an entirely new area of human activity the most authoritative way of proceeding would be through a common appreciation of the problems and common agreement on its solution. The principle of consensus was adopted as the only acceptable procedure for obtaining workable effective solutions.[5]

Conflicting positions must be resolved by unanimous agreement or, if not unanimous agreement, by the dissenting member noting or reserving its objections or position on the record. Of course, each member State, whether on the Outer Space Committee or not, can later decide if it wishes to become a party to the treaty and be bound by its terms. In general, the space treaties have received widespread, though not universal, acceptance by the international community. The Outer Space Treaty, for example, has been signed by 90 countries, and ratified by 56.[6]

The consensus principle has worked well to date, and its usefulness was recently reinforced by the approval in the last session of the Committee of the Agreement Governing Activities of States on the Moon and Other Celestial Bodies.[7] It has worked

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[5]*Supra* note 1.

[6]U.S. Senate Comm. on Commerce, Science, and Transportation: Space Law, 95th Cong., 2d Sess. 36 (2nd ed., Comm. Print, 1978).

[7]Agreement Governing Activities of States on the Moon and Other Celestial Bodies, U.N. GAOR, Supplt. 20, Doc. A/3420 (1979), [*hereinafter* cited as Moon Treaty]. For a text, see Current Documents in this issue of the Journal of Space Law *cf.*

notwithstanding that Committee membership consists of countries with differing systems of jurisprudence, political philosophies, social customs, and levels of scientific and economic development. The process has worked probably because nations have perceived, if not a unity of interests, at least roughly parallel interests in securing outer space for the peaceful pursuit of its potential benefit for all nations and mankind.

III. Accomplishments

What has the Committee on the Peaceful Uses of Outer Space and it subsidiary bodies actually accomplished in 20 years? Taking note of the problems and limitations which inhere in its methods of operations and given the fact that this new organization was dealing not only with the new milieu of outer space but a new frontier of law with virtually no precedent, the number of treaties and conventions presently in force dealing with outer space is surprising, especially when one considers the intricacies of international law and diplomacy and the fact that many disparate interests are involved in outer space. In the scientific and technical area, it has fostered international cooperation and provided a framework for the discussion and dissemination of the results of space activities.

The early work of the Committee, although brought to a halt several times as the result of cold-war tensions, eventuated in General Assembly Resolution 1721 (XVI) in December 1961 which commended to States the following guiding principles: (a) that international law, including the Charter of the United Nations, applies in outer space; and (b) that outer space, including celestial bodies, is free for exploration and use by all States in conformity with international law and is not subject to national appropriations.[8] Six years later, these principles became the main pillars of the 1967 Outer Space Treaty.[9]

In the short span from 1958 to 1974, the Legal Subcommittee has produced four space treaties which have entered into force: The 1967 Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies;[10] The 1968 Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space,[11] the 1972

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[8]U.N.G.A. Res. 1721 (XVI) of December 20, 1961.

[9]Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, January 27, 1967 *(hereinafter* the Outer Space Treaty), [1967] 18 U.S.T. 2410, T.I.A.S. 6347, 610 U.N.T.S. 205 (effective October 10, 1967.

[10]*Ibid.*

[11]The Agreement on the Rescue of Astronauts, the Return of Astronauts, and the Return of Objects Launched Into Outer Space, [1968] 19 U.S.T. 7570, T.I.A.S. 6599, 672 U.N.T.S. 119 (effective December 3, 1968).

Convention on International Liability for Damage Caused by Space Objects,[12] and the 1974 Convention on Registration of Objects Launched into Outer Space.[13]

It is generally accepted that the Outer Space Treaty of 1967 is the basic charter or constitution governing space activities. The extensive history behind the Outer Space Treaty can be inferred from the lapse of six years from Resolution 1721. The Outer Space Treaty includes articles establishing a number of basic principles, such as State responsibility for damage caused by the launching of space objects,[14] provision for the rescue and return of astronauts and objects launched into space,[15] and provision for the registration of space objects so as to identify ownership of the object.[16] The three treaties that followed elaborated these basic principles first established in the Outer Space Treaty. Other articles in the Outer Space Treaty provide that, unlike airspace, outer space, including the Moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty;[17] outer space, including the Moon and other celestial bodies, shall be free for exploration and use by all States without discrimination of any kind;[18] the placing in orbit around the earth, stationing in outer space, or the placement on celestial bodies of any objects carrying nuclear weapons or any other kinds of weapons of mass destruction is proscribed;[19] the Moon and other celestial bodies shall be used exclusively for peaceful purposes;[20] States parties to the Treaty bear international responsibility for national activities including activities of nongovernmental entities in outer space, including the Moon and other celestial bodies;[21] and several provisions providing that space activities should be conducted so as to promote international cooperation in the exploration and use of outer space, including the Moon and other celestial bodies.[22]

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[12]Convention on International Liability for Damage Caused by Space Objects, March 29, 1972, [1973] 24 U.S.T. 2389, T.I.A.S. 7762 (effective October 9, 1973).

[13]Convention on Registration of Objects Launched into Outer Space, January 14, 1979, [1976] T.I.A.S. 7762 (effective September 15, 1976).

[14]Outer Space Treaty, Arts. VI and VII.

[15]*Id.* Art. V.

[16]*Id.* Art. VIII.

[17]*Id.*, Art. II.

[18]*Id.*, Art. I.

[19]*Id.*, Art. IV, para. 1.

[20]*Id.*, Art. IV., para. 2.

[21]*Id.*, Art. VI.

[22]*Id.*, Arts. I, III, IX, X, XI, and XIII.

IV. The Moon Treaty

At its recent 1979 session, the COPUOS reached consensus on a fifth treaty, the Agreement Governing Activities of States on the Moon, and Other Celestial Bodies.[23] This treaty, like its three predecessors, is an elaboration of basic principles first established in 1967 Outer Space Treaty. In 1971, the U.S.S.R. proposed for inclusion in the agenda of the twenty-sixth session of the General Assembly an item entitled "Preparation of an international treaty concerning the Moon."[24] On November 5, 1971, at the twenty-sixth session of the First Committee of the General Assembly, the U.S.S.R. submitted a "Draft Treaty Concerning the Moon."[25] On November 29, 1971, the General Assembly adopted resolution 2779 (XXVI) in which it took note of the draft treaty submitted by the U.S.S.R. and requested the COPUOS and its Legal Subcommittee to consider, as a matter of priority, the question of the elaboration of a draft international treaty concerning the Moon and to report thereon to the General Assembly at its twenty-seventh session.[26]

During the eleventh session of the Legal Subcommittee, a substantially revised and expanded text consisting of a preamble and twenty-one articles was approved as the basis on which work should be pursued during the following session of the Legal Subcommittee as a matter of priority.[27] In the years following 1972, it soon became apparent that there were only three outstanding issues preventing consensus; the scope of the Treaty, *i.e.*, whether it should be restricted to the Moon or apply to other celestial bodies; information to be furnished; and exploitation of natural resources.

During the seventeenth session of the Legal Subcommittee (March 13-April 7, 1978), a concerted effort by the Austrian delegation to resolve the outstanding issues through the medium of informal consultations resulted in a draft text of an "Agreement Governing the Activities of States on the Moon and Other Celestial Bodies" being submitted to the Legal Subcommittee as a working group paper.[28] Because the draft had been circulated so close to the end of the session, neither the working group, the Legal Subcommittee nor the COPUOS had the opportunity to either review or discuss this draft text. Action on the text was deferred to the 1979 session of the Legal Subcommittee so as to permit review and consideration by the delegates and their respective governments. The informal consultations held during the

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[23]Moon Treaty, *supra*, note 7.

[24]U.N. Doc. A/C.1/L.568 (Nov. 1971).

[25]*Ibid.*

[26]U.N. Res. 2779 (XXVI) of November 29, 1971.

[27]U.N. Doc. A/AC.105/101.

[28]U.N. Doc. A/AC.105/218, Annex I, Appendix, WG.I (1978)/WP.2 of 3 April 1978.

1978 Legal Subcommittee and COPUOS sessions concentrated on the natural resources article and, in particular, the inclusion of "the common heritage of mankind" concept, it being informally understood the other outstanding issues of information and scope of treaty could be resolved if the natural resources issues could be successfully compromised.

During the eighteenth session of the Legal Subcommittee (March 12-April 6, 1979), the Austrian draft text was reviewed and discussed.[29] Brazil, speaking for the developing countries, proposed one amendment to the natural resources article (Article XI) which, if acceptable to the rest of the Legal Subcommittee, would be referred to their Governments for acceptance. It was proposed that the "common heritage of mankind" concept be reformulated to read:

> "The moon and its natural resources are the common heritage of mankind, which finds its expression in the provisions of this Agreement and in particular in paragraph 5 of this article."[30]

This resulted in brackets being placed by other delegations around other provisions, *i.e.*, treaty scope, advance notification of placement of radioactive materials, natural resources exploitation review conference, and ratification.

The COPUOS, at its 1979 session, was able to reach consensus due to the acceptance by the U.S.S.R. of the Brazilian formulation of the "common heritage" principle and the agreement of the developing countries not to insist on a provision imposing a moratorium on the exploitation of natural resources pending the establishment of an international regime to govern such exploitation.[31]

Before discussing in greater depth the natural resources Article XI and some other issues of importance in the Treaty, it is noted that the Treaty adds to or usefully clarifies existing international space law in the following respects:

a. It usefully details certain specific prohibitions to ensure peaceful use of the Moon and other celestial bodies;[32]

b. It provides for notification and specified information upon completion of a mission or, for a mission of long duration, every 30 days; provisions in regard to multiple use of areas or orbits, possible discovery of organic life, and danger to human life;[33]

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[29]U.N. Doc. A/AC.105/240.

[30]*Id.* Annex III.

[31]U.N. GAOR, Doc. A/34/20, at 11 (1979).

[32]Moon Treaty, Art. III, para. 2.

[33]*Id.*, Art. V.

c. It confirms right of States to collect and remove samples for scientific investigation and the right to use appropriate quantities of substances on the Moon and other celestial bodies to support scientific missions;[34]

d. It provides for environmental protections;[35]

e. It stipulates advance notification of radioactive materials' use;[36]

f. It requires reporting of areas of special scientific interest and designation of such areas as international scientific preserves;[37]

g. It enunciates the principle of using only such area of the Moon and other celestial bodies as is required for the needs of a station;[38]

h. It imposes the obligation to safeguard human life and to regard all persons on the Moon and other celestial bodies as entitled to Astronaut Agreement treatment;[39]

i. It imposes the obligation to offer shelter to those in distress;[40]

j. It assures the right to humanitarian emergency use of lunar facilities of others;[41]

k. It provides for notification regarding crash landings;[42]

l. It recognizes that a detailed liability regime may be necessary;[43]

m. It provides for the establishment of principles calling for orderly and safe development and rational management of natural resources, expansion of opportunities in use of such resources, equitable sharing in benefits, and special consideration of the interests and needs of developing countries and for the efforts of space powers;[44]

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[34]*Id.* Art. VI, para. 2.

[35]*Id.* Art. VII, para. 1.

[36]*Id.* Art. VII, para. 2.

[37]*Id.*, Art. VII, para. 3.

[38]*Id.*, Art. IX, para. 1.

[39]*Id.*, Art. X., para. 1.

[40]*Id.*, Art. X., para. 2.

[41]*Id.*, Art. XII., para. 3.

[42]*Id.*, Art. XIII.

[43]*Id.*, Art. XIV, para. 2.

[44]*Id.*, Art. XI, para. 7.

n. It details provision for effective consultations in disputes situations.[45]

Coming back to the "common heritage of mankind" principle which is enunciated in Article XI, paragraph 1, and is tied to the future international regime that may be established to govern the exploitation of natural resources, it was the United States which introduced this language in 1972.[46] It confirms that all the States parties to the treaty have a sufficient interest in the possible future exploitation of the natural resources of the Moon and other celestial bodies, that their views are to be given serious consideration at any future international conference which may attempt to establish an international regime. The United States, in an on-the-record statement by the U.S. Representative at the Legal Subcommittee on May 3, 1972, stated the following:

> "On the broadest level of generality, it seems right to state that such resources are of 'the common heritage of all mankind. . . .' On the other hand we would not want to preclude in any way the use of natural resources of celestial bodies for scientific investigation; U.S. activities in returning lunar samples and in sharing them with scientific institutions around the world are well known, as are the Soviet Lunokhod returns and exchanges. We would also want to be careful to ensure that celestial body resources may be used where found for supporting life systems as, for example, in uses by astronauts of liquids or gases of a particular celestial body. Finally, we would need to contemplate a special treaty-drafting conference in the event of the discovery of commercially exploitable resources. At such a conference participants would need to bear in mind not only common goals of economic advancement but the need to encourage investment and efficient development as well."[47]

A further explanation of the meaning attached to the "common heritage" principle by the United States was presented to the Legal Subcommittee by the U.S. Representative on April 19, 1973, as follows:

> "My delegation wants also to place on record United States views concerning the problem of celestial body natural resources. As we have seen the matter, what is involved is not so much a problem but only a conceivable possibility. Essentially, the question is what rules and procedures should reasonably apply in the event of the future discovery of exploitable natural resources on, for example, the Moon.
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> "Last spring the United States proposed that, and I quote, 'The natural resources of the Moon and other celestial bodies shall be the common heritage of all mankind. . . We also proposed to protect current or possible future uses of these resources for scientific research, such as the return of lunar rocks, for example, the hypothetical use by an astronaut of water or some other liquid found in rocks on a celestial body. In addition, we proposed that if and when practical exploitation of lunar resources should become a reality, the parties to the treaty should join in an international conference, and again I

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[45] *Id.*, Art. XV, paras. 2 and 3.

[46] U.N. Doc. A/AC.105/C.2 (XI) WP 12 (April 13, 1972).

[47] U.S. Representative's statement before the Legal Subcommittee of UNCOPUOS, May 3, 1972. For the Report of the Subcommittee, U.N. Doc. A/AC.105/101 (1972).

quote, 'with a view to negotiating arrangements for the international sharing of the benefits of such utilization.'

"As far as they go, these proposals have met with very wide acceptance. When there was some reluctance to proclaim the current applicability of the 'common heritage' principle, we agreed that the Moon Treaty might instead put the matter in a somewhat different context by providing that the purpose of such a future conference should be 'on the basis' of the common heritage principle. On the other hand, as I have at this session repeatedly, although I hope politely, made clear, the United States is not prepared to accept an express or implied prohibition on the exploitation of possible natural resources before the international conference meets and agrees on appropriate machinery and procedures and a treaty containing them takes effect. In our view, the Moon agreement cannot reasonably seek to require that exploitation must await the establishment of the treaty-based regime.

"We have sought to meet concerns of other delegations by accepting the suggestion that the current treaty should call for States conducting missions to the Moon and other celestial bodies to report not only on scientific results but on natural resources. A provision of this character would help ensure that all parties to the treaty would be informed and could take action to prepare for and meet in the international conference.

"One or two particular points should be made concerning these matters as they are reflected in Working Paper 15 which the United States delegation introduced on April 17. As is apparent from the text, this working paper excludes the concept of a pre-regime moratorium. References to the words 'in place' in the first sentence of that paragraph and to paragraph 7 of Article X make this clear. More particularly, the words 'in place' in the first sentence of paragraph 2 are intended to indicate that the prohibition against assertion of property rights would not apply to natural resources once reduced to possession through exploitation either in the pre-regime period or, subject to the rules and procedures that a regime would constitute, following the establishment of the regime. Also with regard to the last sentence of paragraph 2 of Article X, the 'without prejudice' clause would apply to exploitation whether by a State, government entity, nongovernmental enterprise or international organization."[48]

These statements by the United States were not contradicted and constitute a part of the legislative history of the treaty negotiations. (The phrase "in place," referred to in the above statement, is contained in the first sentence of paragraph 3, Article XI, of the Moon Treaty text.)

On the last day of the 1979 COPUOS session, July 3, 1979, after consensus was reached on the Moon Treaty, I, as the U.S. Representative, made the following statement which was again not contradicted by any of the delegations which spoke subsequently:

"Article XI of the draft Moon agreement, which declares that celestial bodies other than the Earth, and the natural resources of such celestial bodies, are the common heritage of mankind, was initially suggested by Argentina but was formally proposed by my delegation in 1972. It makes clear that the parties to the agreement undertake, as the

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[48]U.S. Representative's statement before the Legal Subcommittee of UNCODUOS, April 19, 1973. For the Report of the Subcommittee, see U.N. Doc. A/AC.105/115 (1973).

exploitation of the natural resources of the celestial bodies other than the Earth is about to become feasible, to convene a conference to negotiate an international regime to govern the exploitation of those mineral and other substantive resources which may be found on the surface or subsurface of a celestial body. The draft agreement—and I am particularly pleased about this, as a member of the National Aeronautics and Space Administration (NASA)—as part of the compromises made by many delegations, places no moratorium upon the exploitation of the natural resources on celestial bodies, pending the establishment of an international regime. This permits orderly attempts to establish that such exploitation is in fact feasible and practicable, by making possible experimental beginnings and, then, pilot operations, a process by which we believe we can learn if it will be practicable and feasible to exploit the mineral resources of such celestial bodies. My Government will, when and if these negotiations for such a regime are called for, under Articles XI and XVIII, make every effort to see that the regime is successfully negotiated.

"We note also with satisfaction that Article XI, paragraph 8, by referring to Article VI, paragraph 2, makes it clear that the right to collect samples of natural resources is not infringed upon and that there is no limit upon the right of States parties to utilize, in the course of scientific investigations, such quanitites of those natural resources found on celestial bodies as are appropriate for the support of their missions. We believe that this, in combination with the experimental and pilot programmes, will foster and further, and perhaps speed up, the possibility of the commercial or practical exploitation of natural resources."[49]

In connection with the "common heritage" principle, the 1979 COPUOS Report will record the Committee's agreement that "by virtue of Article I, paragraph 1, the principle contained in Article XI, paragraph 1, would also apply to celestial bodies in the solar system other than the Earth and to its natural resources."[50] The plain meaning of this Committee agreement is to limit application of the "common heritage" principle to the celestial bodies themselves and to the natural resources of such celestial bodies. Clearly, there is no intent to apply the principle to orbits and trajectories of space objects. Further, nothing in the text suggests that all countries are to share equally in the Moon's resources. Any sharing of resources would have to be agreed to in an international conference. Article XI, paragraph 7, uses the phrase "equitable," not "equal" sharing. In determining "equitable" sharing, special consideration is to be given not only to the needs and interests of the developing countries but also to "the efforts of those countries which have contributed either directly or indirectly to the exploration of the Moon."[51]

Regarding Article I, the 1979 COPUOS report will also record the Committee's agreement that "the trajectories and orbits mentioned in Article I, paragraph 2, do not

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[49]U.N. Doc. A/AC.105/PV.203 (July 16, 1979).

[50]*Supra* note 31, at 11.

[51]*Id.* Moon Treaty, Art. XI, para. (d).

include trajectories and orbits of space objects between the earth and such orbits."[52] In my July 3, 1979, statement for the record, I affirmed the U.S. understanding of the Article's meaning as follows:

> "We accept the Committee's conclusions as to this Article—namely, first, that references to the Moon are intended to be references also to other celestial bodies, other than the Earth; secondly, that references to the Moon's natural resources are intended to comprehend those natural resources to be found on these celestial bodies; and thirdly, that the trajectories and orbits referred to in Article I, paragraph 2, do not include trajectories and orbits of space objects in Earth orbit only, or trajectories of space objects between the Earth and Earth orbit.
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> "In regard to the phrase 'Earth orbit only,' the fact that a space object in Earth orbit also is in orbit around the Sun does not bring the space objects which are only in Earth orbit within the scope of this treaty; and a space object orbiting the Moon, while the Moon orbits the Earth as well as the Sun, is in fact within the scope of this treaty."[53]

The "Agreement Governing Activities of States on the Moon and Other Celestial Bodies" will be forwarded to the Special Political Committee of the United Nations General Assembly which will draft a resolution to be submitted to the General Assembly. If adopted by the General Assembly, the treaty will be commended to member States for signature and ratification.

V. Second U.N. Conference on the Peaceful Uses of Outer Space

Moving now to the second United Nations Conference on the Exploration and Peaceful Uses of Outer Space, the Committee agreed that the conference would be held in the latter part of 1982 for a period of two to three weeks and deferred the venue question to 1980.[54] The agenda would permit discussion of "scientific, technical, social, economic, organizational and other relevant aspects and their interrelationships." Attempts to add "legal aspects" were made by several delegations, but they were persuaded to withdraw their proposal by arguments that to include "legal aspects" would change the character of the conference from a scientific and technically oriented meeting to one of political debate, thus possibly frustrating the major purpose for which the conference was designed.[55]

The Committee also agreed that the consensus procedure is to be included in the Rules of Procedure for the Conference to be established, as follows:

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[52]*Supra* note 31, at 11.

[53]*Supra* note 49.

[54]See *supra* note 31, at 22.

[55]*Id.* at 16.

> "It is proposed that best endeavors would be made to ensure that the work of the conference and the adoption of its final report would be accomplished by general agreement."[56]

This language is similar to that pursuant to which the COPUOUS itself operates by consensus. It was recommended that a conference secretariat be established, headed by a secretary-general and three deputies, as well as a conference bureau headed by a president, several vice presidents and chairmen of the three main committees: (1) State of Space Science and Technology; (2) Applications of Space Science and Technology; and (3) International Cooperation and the Role of the UN. If the proposed agenda, which was developed in the Scientific and Technical Subcommittee and approved by the full Committee, is adopted by the UN General Assembly, it will be circulated to all member States in January 1980 with an invitation to submit national papers to be reviewed by the conference secretary-general by spring 1981.[57]

The Committee endorsed the recommendations of the Scientific and Technical Subcommittee that the agenda for its 1980 session should include the following priority items:

> "(a) Consideration of the United Nations Programme on Space Applications and the co-ordination of space activities within the United Nations system;
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> (b) Questions relating to remote sensing of the Earth by satellites;
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> (c) Use of nuclear power sources in outer space;
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> (d) Co-ordinating role of the United Nations in the use of space science technology, particularly in the developing countries.[58]
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> "With regard to the item on remote sensing, the Subcommittee recommended that its future examination of questions relating to remote sensing also include the specific areas of applications of remote sensing technology so as to enable it to further assess the needs of Member States, particularly the developing countries, in the various areas of current and future applications."[59]

It endorsed the recommendation that the agenda of the Scientific and Technical Subcommittee's seventeenth session should also include the following items:

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[56] *Id.* at 20.

[57] *Id.* at 18.

[58] *Id.* at 22. For a report of the Scientific and Technical Subcommittee, See U.N. Doc. A/AC.105/238 (1979).

[59] *Supra* note 31, at 22.

(a) Questions relating to space transportation systems;

(b) Examination of the physical nature and technical attributes of the geostationary orbit.[60]

The Committee further endorsed the Scientific and Technical Subcommittee's recommendation that the Working Group (WG) on Nuclear Power Sources should meet during the session in order to continue its consideration of questions related to the use of nuclear power sources in outer space.[61] The WG has recommended that to assist its future work further studies should be made in the following subject areas:

(1) Elaboration of an inventory of the safety problems involved in the use of nuclear power sources in outer space; (2) Implementation of the recommendations of the International Commission on Radiological Protection (ICRP) relating to populations and the environment in the context of space vehicles utilizing nuclear power sources; (3) Evaluation of existing methods for understanding orbital mechanics to determine if improvements may be made in predicting re-entry phenomena; (4) Definition of technical considerations with regard to a format for notification.[62]

To this end the UN Secretary General has requested member States and international organizations to contribute by September 1979 studies on the technical aspects and safety measures relating to the use of nuclear power sources in outer space, including the above-mentioned aspects identified by the WG as requiring further examination.[63] An informal meeting of technical experts to discuss these papers is expected later this year in preparation for the 1980 Scientific and Technical Subcommittee session.

VI. Legal Subcommittee Agenda for 1980

Regarding the 1980 agenda of the Legal Subcommittee, the Committee recommended that the Legal Subcommittee should continue its work on items pertinent to:

(a) Legal implications of remote sensing of the Earth from space, with the aim of formulating draft principles;

(b) Elaboration of draft principles governing the use by States of artificial earth satellites for direct television broadcasting. . . .

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[60]*Ibid.*

[61]*Supra* note 31, at 9.

[62]*Ibid.*

[63]Note from United Nations Secretary General to U.S. Permanent Representative to the United Nations, dated March 29, 1979.

> [(c)] . . . Matters relating to the definition and/or delimitation of outer space and outer space activities, bearing in mind, *inter alia*, questions relating to the geostationary orbit . . . . [The Committee further recommended] that the Legal Subcommittee should continue to include on its agenda an item entitled 'Other matters.'[64]

The subject of nuclear power sources was added as a separate item to the agenda of the Legal Subcommittee under the title, "Review of existing international law relevant to outer space activities with a view of determining the appropriateness of supplementing such law with provisions relating to the use of nuclear power sources in outer space."[65] In this connection, at the suggestion of the U.S. Representative, the UN will request views of States on the question of relevant international law to be submitted by December 15, 1979.[66] The UN will circulate such submitted views by February 15, 1980.

VII. Remote Sensing

As to remote sensing and direct television broadcasting from satellite, the Committee did not go beyond the work accomplished at the 1979 session of the Legal Subcommittee. There are seventeen (17) draft principles dealing with remote sensing under discussion in the Legal Subcommittee.[67] Because of space limitations, I will only discuss those that form the center of controversy.

While much redrafting of the principles remains to be done, there is one unresolved key issue reflected in a number of the draft principles on which a diversity of views still exists despite extensive discussion over the last four or five years. The issue is access to and the dissemination of remote sensing data and/or information derived therefrom. In very early discussions, questions were raised about the right of a sensing State to sense the territory of another State without the latter's prior consent.[68] With more and more States having this capability and recognizing the potential value of this space application, this argument has largely disappeared and in its stead many delegations now seek to impose a prior consent regime on the dissemination of data and/or information.[69] Other delegations, including the United States, support the right of a sensing State or of a receiving ground station State to openly disseminate remote

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[64]*Supra* note 31, at 22 and 23.

[65]*Id.* at 23.

[66]*Id.* at 10.

[67]*Supra* note 29, Annex I, Appendix A.

[68]U.N. Doc. A/AC.105/147, Annex III.

[69]*Supra* note 29, Annex I.

sensing data without the prior consent of the sensed State.[70] The USSR has proposed that data having a resolution no better than 50 meters photographic be openly disseminated while data better than 50 meters resolution be subject to the prior consent of the sensed State.[71]

Those delegations desiring a prior consent regime argue that remote sensing data and information derived therefrom have economic, political and national security implications. The U.S. delegation has asked delegates for the past three years to cite a single example where Landsat data has damaged the economic, political or national security interest of any State. None has been forthcoming. There are numerous examples where it has benefited States, particularly the developing countries. Benefits usually have some burdens associated with them. The U.S. believes that Landsat-type systems can continue to provide significant national, regional, and international benefits that far exceed the slight risk of injury to the economic, political or national security of any one or more countries.

In addition, if a prior consent regime were adopted, the international cooperative program under which the Landsat ground stations abroad have come into being would have to be ended since data is not only in the possession of the space segment countries but also in the hands of the ground receiving station countries. In response to the Soviet proposal, the U.S. and other delegations pointed out that spatial resolution from a technical standpoint was not a reliable or standard reference, a conclusion supported by the Scientific and Technical Subcommittee as well as by COSPAR, the Committee on Space Research of the International Council of Scientific Unions.[72]

VIII. Direct Broadcast Satellite (DBS)

Regarding the status of the agenda item on elaboration of draft principles governing the use by States of artificial earth satellites for direct television broadcasting a short explanation of what the principles are intended to cover is required. The principles are intended to apply to satellites which will transmit a TV signal from space directly into a home TV receiver. The technology capable of doing this exists today. Japan with the launch assistance of NASA has deployed such an experimental DBS,[73] and a consortium of Nordic countries is planning one for the early 1980s.[74] In order to receive the TV signal, a dish-type antenna approximately one meter and a converter attached to a standard TV set are necessary. In addition, the antenna has to be accurately

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[70]*Ibid.*

[71]*Id.* at Annex I, Appendix B, WG III (1979)/WP1/Rev. 1.

[72]U.N. Doc. A/AC.105/164.

[73]Launched by NASA using Delta vehicle, April 7, 1978.

[74]U.N. Doc. A/AC.105/PV.193.

pointed at the satellite stationed in geostationary orbit. This item has been on the agenda since 1972 and has been the subject of intense debate since completion of the Registration Convention in 1974.[75] A draft preamble and 12 draft principles have been elaborated,[76] but no consensus has been reached though a great deal of time and effort has been expended looking for a compromise solution.

There are in essence only two issues which appear to be blocking consensus, and it is on these issues that various compromise formulations have been proposed. These issues relate to the formulation of the principle on consultation and agreement between States. Both involve the concept of prior consent. But before discussing the issue, some additional background is needed.

In 1977, a World Administrative Radio Conference was held under the auspices of the I.T.U. and a result of the conference was the adoption of a plan consiting of national assignments of frequencies and orbital slots for direct television satellite broadcasting for Regions 1 and 3, which include the world except for North and South America in Region 2.[77]

It would be a breach of the plan adopted at the 1977 conference for a country that has agreed to adhere to the plan to use in Regions 1 and 3 an orbital position or frequency channel not assigned to it in order to broadcast to another country in those regions. The 1977 conference also treated the question of spillover. Broadcast satellites transmit a wide beam. The beam is shaped by the antennas aboard the spacecraft, but it is impossible to shape the beam to conform with national boundaries and so you have what is called "spillover" into neighboring countries. The 1977 conference established limits on this unavoidable spillover, seeking to reduce it to a minimum.[78]

With this background, I can proceed to the issues. The U.S.S.R. and the Eastern Bloc countries insist, notwithstanding the I.T.U. regulations which because of the need for rational management of the radio frequency have imposed by technical regulations what is in effect a prior consent regime for direct broadcast satellites in Regions 1 and 3, that there ought to be UN/DBS principles which establish politically a requirement that direct broadcast signals cannot cross national borders, whether intentional or unintentional, without the prior consent and agreement of the neighboring State.[79] A large group of delegations would exclude the requirement for prior consent in the case of unintentional spillover but would apply it to the case where a State intends to

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[75] *Supra* note 13.

[76] *Supra* note 29, Annex II, Appendix A.

[77] 1977 World Administrative Radio Conference.

[78] *Ibid.*

[79] *Supra* note 29, Annex II.

broadcast its signal into some State other than its own.[80] The U.S. and a very small number of other delegations view the present language of the proposed principle on consultation and agreements between States as limiting and eroding the principle of free flow of information, a fundamental human right[81] recognized in such international instruments as the Universal Declaration of Human Rights in Article 19 which declares the right of people "to seek, receive, and impart information and ideas through any media and regardless of frontiers" and reaffirmed in Resolution 33/115 of the General Assembly and most recently in the UNESCO Declaration on the Mass Media.[82] The U.S., fully supports a principle providing for full consultations prior to the establishment of any international DBS by a State, consultations which could include the subject of program content, but not consultations that must end with an explicit agreement or the prior consent of a receiving state before such broadcasting may begin. The issue is therefore ideological and political and may be difficult to resolve. It remains to be seen whether consensus will ever be reached as long as this issue remains.

IX. Definition of Outer Space

On the issue of definition of outer space, the U.S.S.R. introduced at the 1979 COPUOS session a working paper which set forth draft provisions for a UN General Assembly resolution governing both the definition question and the legal status of geostationary satellites' orbital space.[83] Essentially, the Soviets propose: (1) that the region above 100/110 km from sea level is outer space, (2) that space objects of States have the right to fly over the territory of other States at lower altitudes for the purpose of reaching orbit and returning to earth in the launching State's territory, and (3) that the geostationary satellites' orbital space is inseparable from outer space and all relevant provisions of the 1967 Outer Space Treaty are applicable to it, *i.e.*, it is not subject to national appropriation.[84]

While the U.S. agrees with the U.S.S.R. as to its characterization of the geostationary orbit (GSO) as being part of outer space, the U.S. has expressed the view that definition is not necessary at the present time as no delegation has identified any problem that would be solved by adopting a definition.[85] Further, activities in space have been going on for over 20 years without hindrance or problems caused by lack of a demarcation line. It may also be noted that COSPAR in one of its reports indicated that

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[80]*Ibid.*

[81]*Ibid.*

[82]UNESCO Decision 4/9.3/2, adopted November 22, 1978.

[83]*Supra* note 7, at 8.

[84]U.N. Doc. A/AC.105/L.112.

[85]*Supra* note 29, at 9.

past estimates of the lowest altitude at which satellites had survived had been too high, and this argues against adopting an arbitrary altitude at this time.[86] At this point no drafting of either principles or a Treaty document on these matters has started.

Within the UN Committee on the Peaceful Uses of Outer Space and its two subcommittees, several equatorial States have asserted that segments of the geostationary orbit are subject to the sovereignty of the underlying States located on the equator.[87] The argument made in support of the claims of sovereignty over the geostationary orbit by the equatorial States is based upon the assertion that the geostationary orbit is not part of "outer space" because the existence of the orbit depends exclusively upon the gravitational phenomena caused by the territory of the underlying States.[88]

Such an argument has no scientific or technical basis. The Scientific and Technical Committee had earlier concluded that there were no scientific or technical characteristics of the earth's upper atmosphere that would be a basis for a definition/delimitation and had requested the Legal Subcommittee to identify problems to be solved by such definition. The geostationary orbit is but one of an infinite number of orbits into which satellites can be placed, and its particular characteristics are functions of both the total gravitational field of the Earth and the rotation of the whole Earth. There is no relationship between the GSO and any underlying country. Entry into the GSO can be achieved through launch from any country on Earth, though energy requirements for insertion into the GSO vary with payload mass and latitude of launch among other factors. The orbit is completely unaffected by national boundaries on Earth, and satellites require adjustment maneuvers to maintain a steady GSO. Further, the GSO is roughly 35,000 kilometers above the Earth's surface, far beyond the altitudes of most satellites currently in Earth orbit. GSOs were understood and utilized before and during the negotiation of the 1967 Outer Space Treaty, and there is no basis for treating the GSO differently than any other Earth orbit. The provisions of Article II of the 1967 Outer Space Treaty preclude any claims of national sovereignty over geostationary orbits or portions thereof.

For all these reasons, I disagree with the recommendation of the panel which in the International Space Activities report of the Subcommittee on Space Science and Applications of the House Committee on Science and Technology urged that "action should be initiated and vigorously pursued to establish an international codicil to the 1967 Outer Space Treaty which specifically forbids any individual nation's claiming

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[86] *Supra* note 72.

[87] *Supra* note 29, at 9.

[88] U.N. Doc. A/AC.105/PV.198, p. 31.

sovereignty over the geostationary''[89] since such a codicil is unnessary as the legal status of the GSO is well established both by treaty and customary international law. Further, there is no reason to believe that the equatorial States would become signatories to such a codicil and abate their claims of sovereignty over such orbits. The second recommendation of the panel is a far better solution; this is to seek ways of reducing through technological developments and improvements the demand for geostationary locations so as to meet the future needs of the international community, and in particular those of the developing countries, including, if feasible, the development of multipurpose space platforms.[90]

It is my belief that there is no need to add a codicil to the 1967 Outer Space Treaty since it is absolutely clear that under that treaty the geostationary orbit is in outer space and is not subject to national appropriation by any means. This is the position of the U.S., U.S.S.R., and the overwhelming majority of member States of the Committee on the Peaceful Uses of Outer Space.

X. Conclusion

As reviewed above, it appears that the 1980 sessions of the COPUOS and its two subcommittees will be dealing with a number of important issues that are difficult to resolve. However, the history of the COPUOS demonstrates that it is a dynamic body with an enviable record of achievement, and there is no reason to expect that the Committee on the Peaceful Uses of Outer Space will not move forward on the matters under consideration.

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[89]Subcomm. on Space Science and Applications of the Comm. on Science and Technology: U.S. House Report on International Space Activities (Nov. 1978).

[90]*Ibid.*